



COSC2671

Social Media and Network Analysis

ASSESSMENT 2 – Group Project Worklog

Who Drives Open-Source AI?

Network Centrality and Discourse Analysis on GitHub ML Repositories

PG-17

John Robyn De Guzman s3944714

Phavan Kuumar Janakiraman Suresh s4190557

Worklog

Week	Dates	Task	Assigned	Status
Week 5	31 Mar – 6 Apr	Assignment 2 briefing review and topic brainstorming	Both	Completed
Week 5	31 Mar – 6 Apr	Initial topic selection: Reddit mental health networks	Both	Completed
Week 6	7 – 13 Apr	Reddit data exploration; assess data quality and API limitations	Rob	Completed
Week 6	7 – 13 Apr	Background literature review: network analysis and sentiment analysis	Phavan	Completed
Week 7	14 – 20 Apr	Topic pivot to GitHub ML repositories; finalise research question	Both	Completed
Week 7	14 – 20 Apr	Define 5 target repositories and write project plan	Rob	Completed
Week 7	14 – 20 Apr	Draft background sections 2.1 and 2.2	Phavan	Completed
Week 8	21 – 27 Apr	Build Notebook 1: data collection via PyGitHub 2.x	Rob	Completed
Week 8	21 – 27 Apr	Debug PyGitHub 2.x authentication and rate-limit breaking changes	Rob	Completed
Week 8	21 – 27 Apr	Run overnight data collection (10,000 issues across 5 repositories)	Rob	Completed
Week 8	21 – 27 Apr	Draft methodology section 3.1 Data Collection	Phavan	Completed
Week 9	28 Apr – 4 May	Build Notebook 2: data cleaning, bot removal, credential redaction	Rob	Completed
Week 9	28 Apr – 4 May	Verify cleaned dataset: 42,550 rows, 23 bots removed, 5 credentials redacted	Rob	Completed
Week 9	28 Apr – 4 May	Draft methodology sections 3.2 and 3.3	Phavan	Completed
Week 10	5 – 11 May	Build Notebook 3: network construction, centrality, Louvain, k-core	Rob	Completed
Week 10	5 – 11 May	Generate all network outputs: GEXF, pyvis HTML, centrality CSVs, figures	Rob	Completed
Week 10	5 – 11 May	Draft Section 4: network analysis results	Both	Completed

Week	Dates	Task	Assigned	Status
Week 10	5 – 11 May	Draft methodology sections 3.4 and 3.5	Phavan	Completed
Week 11	12 – 18 May	Build Notebook 4: VADER, Mann-Whitney U, BERTopic, Spearman correlation	Rob	Completed
Week 11	12 – 18 May	Generate all NLP outputs: sentiment CSVs, topic info, visualisation figures	Rob	Completed
Week 11	12 – 18 May	Draft Section 5: NLP analysis results	Both	Completed
Week 11	12 – 18 May	Draft Section 6: integrated analysis and interpretation	Rob	Completed
Week 12	19 – 22 May	Draft Sections 7 and 8: discussion, limitations, and conclusion	Both	Completed
Week 12	19 – 22 May	Compile full report; insert all figures, tables, and network comparison image	Rob	Completed
Week 12	19 – 22 May	Proofread and verify all numbers against notebook outputs	Both	Completed
Week 12	19 – 22 May	Write references, self-reflection, and project plan	Both	Completed
Week 12	22 May	Final submission	Both	Completed

Tools and Environment

Tool / Library	Purpose
Python 3.x / Jupyter Notebooks	All data collection, analysis, and visualisation
PyGitHub 2.x	GitHub REST API data collection — authenticated as s3944714
NetworkX, python-louvain	Network construction, PageRank, betweenness centrality, Louvain community detection, k-core decomposition
VADER, BERTopic, SentenceTransformers	Sentiment scoring, topic modelling with all-MiniLM-L6-v2 embeddings
pandas, scipy, matplotlib, seaborn, pyvis	Data processing, statistical tests (Mann-Whitney U, Spearman), visualisation, interactive network HTML
GitHub (username: s3944714)	Version control for all notebooks, raw data tracking, and collaboration

Tool / Library	Purpose
Microsoft Word	Report writing, table formatting, and final document assembly

Self-Reflection: Rob

This project was technically demanding, requiring four interconnected Jupyter notebooks, real API data collection, and postgraduate-level integrated analysis. I took primary responsibility for all technical implementation — data collection, cleaning, network analysis, NLP analysis, and report writing.

The most satisfying outcome was the integrated Spearman correlation analysis in Section 6. Interpreting the consistently negative centrality-sentiment relationship as a governance behaviour signature, rather than a weak result, required genuine analytical thinking that I feel represents real postgraduate-level work.

What worked well was the four-notebook pipeline architecture with clean handoffs between stages, making the workflow reproducible and debuggable. The vectorised edge construction approach also prevented performance issues across 42,550 rows.

What did not work well was the pytorch data collection strategy. Collecting only the 2,000 most recent issues produced a narrow three-month window that caused the SC4 BERTopic failure. In future I would use stratified sampling across the full date range for large repositories. Time management in the final week was also tighter than ideal — I would set a hard notebook-completion deadline five days before submission in future projects.

Self-Reflection: Phavan

This project gave me a deeper understanding of how social network analysis and NLP can be combined to answer meaningful research questions about real-world communities. My primary contributions were the literature review, background section, methodology writing, and the discussion and implications sections of the report.

Working alongside Rob, who handled the full technical pipeline, required clear communication about what the analysis was producing so I could write about it accurately. The iterative process of verifying every reported statistic against the actual notebook outputs before finalising the report was something I found valuable — it reinforced how important it is that written analysis stays grounded in the actual data rather than generalised claims.

What worked well was the division of labour. Separating technical implementation from research writing allowed both components to develop in parallel without bottlenecks. The background section in particular benefited from having dedicated time to engage properly with the literature rather than treating it as secondary to the code.

What I would improve is getting more involved in the analytical interpretation earlier. I found the discussion of the Spearman correlation results — specifically framing the negative centrality-sentiment finding as governance behaviour rather than a weakness — was the most intellectually interesting part of the project, and I would have contributed more to that interpretation with deeper early engagement with the network analysis outputs.

